
Overview:

I am a Ph.D. Candidate at UC San Diego. My research centers around developing safe physical intelligence for robots, specifically tackling the challenges of uncertainty in unstructured real world environments. In particular, I am driven by a passion to develop intelligent systems that can significantly improve quality of life and enhance accessibility for all. Specifically my areas of research are:

- **Adaptable Safety Guarantees via Online Learning:** My research uses non-parametric online learning methods to adapt safety assurances while maintaining probabilistic safety guarantees.
- **Uncertainty Aware Safe Automation:** My work focuses on leveraging uncertainty quantification and probabilistic estimation to give safety guarantees in unknown, unstructured environments.
- **Surgical Automation:** Developing algorithms for task automation specific to minimally invasive surgery.

Key Areas of Expertise and Research:

- Robotics
- Robot Manipulation
- Safe Control
- Learning for Safe Control in Robotics
- Learning Representations for Control, Planning and Safety in Robotics
- Robotic Surgical Automation
- Policy Learning for Robotics
- Online Learning for Robotics

Education

University of California San Diego

Ph.D. Intelligent Systems, Robotics and Control: *GPA 3.95*

Graduation: Feb 2026

NDSEG Fellow, Powell Fellowship

University of California Berkeley

B.S. Electrical Engineering and Computer Science: *GPA: 3.855*

Graduated with Honors: May 2019

Regents and Chancellor's Scholar

Professional Experience:

- **UCSD Advanced Robotics and Controls Lab (ARCLab) and Safe Autonomous Systems Lab (SASLab)** *2020 - Present*
 - Conducting Doctoral research with advisors Michael C. Yip and Sylvia L. Herbert at UC San Diego.
 - Research on surgical automation, safe manipulation in unstructured environments, safe control under uncertainty, robust manipulation, learning for robotics.
 - Published and presented multiple papers on safe control, robotic manipulation and surgical automation in top robotics conferences including RSS, CoRL, ICRA, IROS, L4DC and CASE.
 - Conducted experiments on hardware platforms including a Manipulator, Intuitive Da Vinci Surgical Robot and UAVs.
 - Advised and mentored multiple undergraduate and masters students.
- **UCSD Wireless Communication, Sensing and Networking Lab (WCSNG)** *2019 - 2020*
 - Doctoral research advised by Professor Dinesh Bharadia.
 - Research focused on using Machine Learning to create robust wireless communication systems.
- **UC Berkeley ML4Wireless Center (Undergraduate).** *2018 - 2019*
 - Undergraduate Research in Machine Learning for Wireless communications advised by Professor Anant Sahai.
 - Research focused on using neural networks to learn wireless protocols, with a focus on channel estimation and equalization.
- **Engineering Intern at Intuitive Surgical: (Sunnyvale Office)** *Jan 2022 - May 2022*
 - Developed unsupervised and supervised learning based computer vision algorithms to improve navigation performance for robotic bronchoscopy procedures for Intuitive Surgical's Ion platform.
- **Engineering Intern at Qualcomm: (San Diego Office)** *June 2018 - Aug 2019*
 - Engineered a suite of Python scripts that streamlined the modification, compilation, and decompilation of modem configuration binaries, improving workflow efficiency for the engineering team.

- Earned a QualStar award for outstanding performance as an Intern.
- **Engineering Intern at Qualcomm: (San Jose Office)** *June 2018 - Aug 2018, June 2019 - Aug 2019*
 - Engineered a new feature in C for a DSP chip simulator, enabling the accurate simulation of hardware timers and significantly enhancing the tool's functionality.
- **Development Engineer Intern at Epicor: (San Diego Office)** *June 2016 - Aug 2016*
 - Used iterative linear modeling on large data logs to isolate high impact variables to improve product performance.
- **LSSI Research Intern at The Scripps Research Institute:** *June 2014 - Aug 2014*
 - Investigated the effects of altering a protein's linked glycosylation site on the specificity of oligosaccharyltransferase and the stability of the altered protein.

Teaching

- **Undergraduate Student Instructor at UC Berkeley:** *Jan 2017 - May 2019*
 - I instructed students on electrical engineering, linear algebra and data analytics concepts in discussion sections as well as how to put these concepts to practice in labs via projects, such as creating a voice controlled car. I also created labs, homeworks, exams, course notes and other course content.
- **Undergraduate Student Instructor for EE16A at UC Berkeley:** *June 2017 - August 2017*
 - I instructed students on introductory electrical engineering labs. I lead office hours to help with concepts in electrical engineering and linear algebra.
- **Lab Assistant for EE16B at UC Berkeley:** *Aug 2016 - Dec 2016*
 - Assisted undergraduate students in completing Electrical Engineering labs, culminating in building a voice controlled car.

Awards and Distinctions

- **Best Paper Award:** Best workshop paper at Reflections on Representations and Manipulating Deformable Objects ICRA 2025.
- **NDSEG Fellow:** Department of Defense National Defense Science and Engineering Graduate (NDSEG) Fellowship in 2021.
- **Competition winner at MRSS (Montreal Robotics Summer School):** Attended a week long robotics course at MILA where my team won the competition to localize and navigate a course using a go1 quadruped trained to walk using RL in IsaacGym in 2024.
- **Powell Fellowship:** Fellowship at UC San Diego 2019.
- **QualStar Award at Qualcomm:** Award for outstanding performance as an Intern 2016.
- **Eta Kappa Nu:** Invited Member of the Mu Chapter of the EECS Honor Society at UC Berkeley.
- **Tau Beta Pi:** Invited Member of the California Alpha Chapter of the Engineering Honor Society at UC Berkeley.
- **Regents and Chancellor's Scholarship:** Prestigious undergraduate scholarship at UC Berkeley.

Service

- Advised Masters students: Haixin Jin, Borui Li, Ashish Maknikar, Siddhant Saoji.
- Advised Undergraduate students: Zihang He.
- Outreach Chair of RoboGrads at UCSD: Outreach chair for Graduate Student organization at UC San Diego where I worked on organizing events to build a collaborative research community and inspire younger students through lab tours and events.
- Associate officer of Service Committee in Eta Kappa Nu: Assisted in organizing and developing content for labs and outreach events to encourage children to pursue a career in Electrical Engineering and Computer Science.
- Project Manager at Enabletech: Consulted and helped manage a variety of Engineering projects for a student organization at UC Berkeley that was focused on innovative solutions to assist people with disabilities.

Technical Skills

- Linear Algebra, Statistical Modeling, Deep Learning, Reinforcement Learning, Imitation Learning, Gaussian Processes, Data Analytics, Simulation, Adaptive Learning, Safe Control, Safe Learning.
 - Python, OpenCV, PyTorch, Multi threaded Programming. Very comfortable picking up new programming languages and packages. Basic skills: ROS, MATLAB, R.
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All Publications

1. **Nikhil Uday Shinde**, Dylan Hirsch, Michael C. Yip, Sylvia L. Herbert, “Safe Stochastic Explorer: Enabling Safe Goal Driven Exploration in Stochastic Environments and Safe Interaction with Unknown Objects”, *In Submission*.
2. Kuancheng Wang, Seungho Yeom^{*}, Jinglin Cao^{*}, Yuheng Zhi^{*}, **Nikhil Shinde^{*}**, Michael Yip “Remember what you did?: Learning Behavioral Memories for Partially Observable Object Manipulation”, *In Submission*.
3. Sander Tonkens^{*}, **Nikhil Uday Shinde^{*}**, Azra Begzadić, Michael C. Yip, Jorge Cortès, Sylvia L. Herbert, “From Space to Time: Enabling Adaptive Safety with Learned Value Functions via Disturbance Recasting”, *In Conference on Robot Learning (CoRL) 2025, Workshop paper at Safe and Robust Robot Learning for Operation in the Real World at RSS 2025, Workshop at RSS 2025*.
4. Haixin Jin^{*}, **Nikhil Uday Shinde^{*}**, Soofiyan Atar, Hongzhan Yu, Dylan Hirsch, Sicun Gao, Michael C. Yip, Sylvia L. Herbert “Learning to Nudge: A Scalable Barrier Function Framework for Safe Robot Interaction in Dense Clutter”, *In submission ICRA 2026*.
5. **Nikhil Uday Shinde^{*}**, Xiao Liang^{*}, Fei Liu, Yutong Zhang, Florian Richter, Sylvia Herbert, Michael C. Yip, “JIGGLE: An Active Sensing Framework for Boundary Parameters Estimation in Deformable Surgical Environments”, *In Robotics Science and Systems (RSS) 2024, Best workshop paper at Reflections on Representations and Manipulating Deformable Objects ICRA 2025*.
6. Will Sharpless^{*}, Dylan Hirsch^{*}, Sander Tonkens, **Nikhil Uday Shinde**, Sylvia L. Herbert “Dual-Objective Reinforcement Learning with Novel Hamilton-Jacobi-Bellman Formulations” *In submission at ICLR 2026, Workshop paper at Safe and Robust Robot Learning for Operation in the Real World at RSS 2025, Workshop paper at RSS 2025*.
7. Xiao Liang, Chung-Pang Wang, **Nikhil Uday Shinde**, Fei Liu, Florian Richter, Michael Yip, “MEDiC: Autonomous Surgical Robotic Assistance to Maximizing Exposure for Dissection and Cautery”, *Accepted to IEEE International Conference on Robotics and Automation (ICRA), 2025*.
8. Azra Begzadić^{*}, **Nikhil Uday Shinde^{*}**, Sander Tonkens^{*}, Dylan Hirsch, Kaleb Ugalde, Michael C. Yip, Jorge Cortés, Sylvia Herbert, “Back to Base: Towards Hands-Off Learning via Safe Resets with Reach-Avoid Safety Filters”, *Submitted to Learning for Dynamics and Controls (L4DC), 2025, Workshop paper at RSS 2025*.
9. **Nikhil U. Shinde^{*}**, Zih-Yun Chiu^{*}, Florian Richter, Jason Lim, Yuheng Zhi, Sylvia Herbert, Michael C. Yip, “SURESTEP: An Uncertainty-Aware Trajectory Optimization Framework to Enhance Visual Tool Tracking for Robust Surgical Automation”, *In IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024*.
10. **Nikhil U. Shinde**, Xiao Liang, Florian Richter, Michael C. Yip, “Investigating Low Data, Confidence Aware Image Prediction on Smooth Repetitive Videos using Gaussian Processes”, *In IEEE 19th International Conference on Automation Science and Engineering (CASE), 2024*.
11. **Nikhil U. Shinde**, Jacob Johnson, Sylvia Herbert, Michael C. Yip, “Object-Centric Representations for Interactive Online Learning with Non-Parametric Methods”, *In IEEE 19th International Conference on Automation Science and Engineering (CASE), 2023*.
12. Will Sharpless, **Nikhil Shinde**, Matthew Kim, Yat Tin Chow, Sylvia Herbert, “Koopman-Hopf Hamilton-Jacobi Reachability and Control”, *Arxiv pre-print*.
13. Christopher D'Ambrosia, Florian Richter, Zih-Yun Chiu, **Nikhil Shinde**, Fei Liu, Henrik I. Christensen, Michael C. Yip, “Robust Surgical Tool Tracking with Pixel-based Probabilities for Projected Geometric Primitives”, *In IEEE International Conference on Robotics and Automation (ICRA), 2024*.
14. Elizabeth Peiros, Zih-Yun Chiu, Yuheng Zhi, **Nikhil Shinde**, Michael C. Yip, “Finding biomechanically safe trajectories for robot manipulation of the human body in a search and rescue scenario”, *In IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023*.
15. Vignesh Subramanian, Laura Brink, Nikunj Jain, Kailas Vodrahalli, Akhil Jalan, **Nikhil Shinde**, Anant Sahai, “Some new numeric results concerning the Witsenhausen Counterexample”, *In 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton), 2018*.

* denotes equal contribution

Additional Relevant Projects:

- **Safe Deformable Manipulation with Vision Language Models:** Using vision language models to enable safe manipulation strategies while manipulating complex deformable objects and scenes where safety can be difficult to quantify.
- **Neural Operators for Adaptive Safety:** Training a Neural Operator to learn an environment's safety value function as a function of spatial wind disturbance, enabling a GP-based approach for online disturbance mapping and adaptive safety filter creation.
- **Interpretable Interventions:** Using Sparse Autoencoders and Transcoders to design interpretable features and interventions for Reinforcement Learning and Imitation Learning policies on robotic systems.

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- **Safe Exploration and Manipulation in Unknown Environments with Stochastic Transitions:** Using Gaussian Processes for safe goal driven exploration and object manipulation in unknown environments subject to a stochastic transition function.
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